Topic 41 — Environmental Monitoring (पर्यावरणीय निगरानी)

Covers syllabus (click to expand)

Environmental Indicators | Environmental Monitoring | Environmental Audit (पर्यावरण लेखापरीक्षा)

Sources baked in: CPCB; CGWB; ISO 14001; EPA Form V; SAFAR/IITM; UPPCS PYQs **Weight:** ★★ Medium — BOD; Ringelmann; CGWB≠CPCB; SAFAR; audit≠monitoring **Last verified:** August 2026

Current Affairs (this topic)

- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) / SPCB monitoring network and satellite products are living CA.
- Keep ambient vs emission monitoring distinction clear.

41.1 Environmental Indicators

Identity: PSR framework | BOD/COD/DO/AQI/dB | Bio-indicators | Ringelmann

- **Environmental indicators** turn raw data into decision signals for policy and public warning.
- **PSR model:Pressure** (emissions, resource use) → **State** (ambient quality) → **Response** (treatment capacity, action plans).
- Air indicators: **PM2.5, PM10, SO_2, NO_2, CO, O_3** (and AQI composite 0–500).
- Water indicators: **BOD, COD, DO, pH, coliform**. Higher BOD = more organic pollution; **COD ≥ BOD**; healthy streams need adequate **DO**.
- Noise indicator: **decibel (dB)** (डेसिबल) — logarithmic; CPCB zone limits apply.
- **Ringelmann scale (रिंगेलमैन पैमाना) (0–5)** measures **smoke plume density/opacity**, not water, fog, or noise.
- Bio-indicators: **lichens** (SO_2-sensitive air quality); aquatic macroinvertebrates for clean streams.

PYQ — UPPCS Prelims 2023, Q103

BOD is a standard criterion for—

Options: A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

Show answer

Ans: A

PYQ — UPPCS Prelims 2021, Q34

Ringelmann scale is used to measure density of

Options: A. Smoke B. Polluted water C. Fog D. Noise

Show answer

Ans: A

PYQ — UPPCS Prelims 2025, Q61

Lichens are good pollution indicators; they do **not** grow well in polluted areas → **Only statement 1**.

Indicator	Medium	High value means
PM2.5 / AQI	Air	Worse air
BOD	Water	More organic pollution
DO	Water	Low = stressed aquatic life
dB	Noise	Louder / possible breach
Ringelmann	Smoke	Denser plume

> **Logic:** BOD ≠ blood oximetry. Ringelmann ≠ fog meter.

41.2 Environmental Monitoring

Identity: Networks | CPCB vs CGWB | SAFAR | NCAP | SPCB consents

- **Environmental monitoring** is systematic, repeated measurement of air, water, soil, noise, and related parameters against standards.
- Cycle: sample → analyse → compare with norms → report → enforce/public alert.
- **CPCB** runs national **air** and **surface-water** programmes and sets ambient/effluent standards with SPCBs.
- **NAMP** = manual air stations for long-term trends; **CAAQMS** = continuous real-time ambient air (feeds AQI boards).
- **NWMP** = National Water Quality Monitoring Programme for rivers/lakes.
- **CGWB** monitors **groundwater** levels and quality; publishes the **Annual Groundwater Quality Report** — not CPCB.
- **SAFAR** (IITM) combines air-quality monitoring with weather forecasting; inaugurated by **Dr. Harsh Vardhan** (July 2018).
- **NCAP (2019)** expands monitoring and action in non-attainment cities (including multiple UP cities).
- **SAMEER** app disseminates CPCB AQI; **PARIVESH** tracks EC compliance reporting.
- **UPPCB** issues Consent to Establish/Operate and inspects industries in Uttar Pradesh (उत्तर प्रदेश).

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by:

1. CPCB 2. CWC (केंद्रीय जल आयोग) 3. CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

Show answer

Ans: B

PYQ — UPPCS Prelims 2018, Q84

SAFAR inaugurated (July 2018) by:

Show answer

Dr. Harsh Vardhan

System	Nodal	Medium
NAMP / CAAQMS	CPCB–SPCB	Air
NWMP	CPCB	Surface water
WQMN / GW Quality Report	CGWB	Groundwater
SAFAR	IITM	Air + forecast
Noise monitoring	CPCB–SPCB	Sound

> **Logic:** CPCB ≠ groundwater atlas. CGWB ≠ AQI boards.

41.3 Environmental Audit

Identity: Compliance verification | Form V | ISO 14001 | ≠ monitoring alone

- **Environmental audit** is a systematic, documented check of whether an organisation meets environmental laws, consent conditions, and internal EMS goals.
- Monitoring **collects data**; audit **verifies compliance** and management systems using that data (and inspections).
- Under **EPA Rules, Rule 14**, specified industries file an annual **Environmental Statement (Form V)** with the SPCB — India's statutory disclosure/audit-type tool.
- **ISO 14001** is the voluntary international standard for an **Environmental Management System (EMS)**.
- Post-EC projects submit periodic **compliance reports** (often six-monthly) — audit/compliance track linked to EIA (Topic 14).
- Green audits in institutions review energy, water, waste, and campus ecology (पारिस्थितिकी).
- Consent to Operate renewal typically depends on monitoring results plus inspection/audit findings.

Tool	Role
Form V statement	Annual statutory disclosure to SPCB
ISO 14001	Voluntary EMS certification
EC compliance report	Project clearance conditions
SPCB inspection	Enforcement audit on ground

> **Logic:** Passing an AQI reading ≠ completing an environmental audit.
